The EnvisionBOX Gesture Classification Challenge: A Multi-Corpus Benchmark for Co-Speech Gesture Detection via Privacy-Preserving Kinematic Extraction

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Abstract

Co-speech gesture research is fragmented: laboratories produce richly annotated corpora in isolation, models rarely generalise beyond the corpus they were trained on, and cross-study comparison is structurally impossible without a shared benchmark. The primary barrier to pooling data across institutions is privacy: raw video of human participants cannot be freely shared under GDPR and equivalent regulations. We address both problems together. We present **EnvisionBOX**, the first multi-corpus, multi-language benchmark for co-speech gesture classification, built entirely on anonymised kinematic data extracted by **MaskBench**, an open-source benchmarking framework that processes video into skeletal landmarks—without releasing raw frames, audio, or face mesh. EnvisionBOX aggregates 32,575 annotated gesture clips from seven corpora, four languages, and five interaction settings (narration, conversation, direction-giving, clinical interview, and public presentation), covering 150+ unique speakers. We describe a speaker-independent evaluation protocol and two competitive baselines: a LightGBM model achieves 75.5% cross-validated and 69.5% held-out balanced accuracy; a 1D residual CNN achieves 66.5% and 60.1% respectively. The persistent gap between cross-validated and held-out accuracy demonstrates that cross-corpus gesture detection is a genuinely open challenge. We release training data, an evaluation script, and a starter kit, and invite the MINT community to build the shared benchmark infrastructure that gesture research has lacked.

1 Introduction

Co-speech gesture is a fundamental modality of face-to-face communication (McNeill, 1992; Kendon, 2004). Speakers gesture spontaneously and universally, and their gestures carry semantic, pragmatic, and interpersonal information that complements the spoken signal. Understanding, modelling, and generating gesture in naturalistic interaction is therefore a central problem for multimodal NLP—and a core concern of the MINT research community.

Despite this importance, gesture research lacks the comparative infrastructure that has driven progress in neighbouring fields. In natural language processing, shared tasks—CoNLL named entity recognition (Tjong Kim Sang and De Meulder, 2003), SemEval semantic similarity, WMT machine translation—created the conditions for rapid community progress: common training and test splits, agreed evaluation metrics, and a public record of which systems work and why. Gesture research has none of this. Laboratories collect and annotate corpora independently, using different gesture taxonomies, different recording conditions, and different segmentation criteria. Models trained on one corpus generalise poorly to another, results cannot be directly compared across studies, and the field re-solves the same annotation and evaluation problems repeatedly.

The root cause is privacy. Video recordings of human participants carry identifiable facial, vocal, and behavioural information. GDPR and equivalent regulations in Germany, the Netherlands, China, and elsewhere prevent laboratories from sharing raw recordings across institutional boundaries without extensive ethical review, consent re-negotiation, and data processing agreements. In practice, this means each lab works with its own data in isolation.

Two recent developments make it possible to break this deadlock. First, pose estimation has crossed a practical threshold. Frameworks such as MediaPipe (Lugaresi et al., 2019) extract full-body 3D landmark coordinates from video at real-time speed on commodity hardware. The resulting world landmark sequences—joint positions in metric space, centred on the hips—carry no identifiable facial or vocal information and can be shared without the privacy risks of the original record-